

Abstract No.

WNT6 modulates extracellular vesicle biogenesis and cargo loading in glioblastoma: implications for liquid biopsies

Helena M. Figueiredo^{1,2,}, Eduarda P. Martins^{1,2}, Alexandra Teixeira³, Lorena Diéguez³, Paula Ludovico^{1,2}, Eduarda Sá-Marta⁴, Marta Viana-Pereira^{1,5}, Marta Almeida⁵, Afonso M. Pinto⁴, Bruno M. Costa^{1,2}, and Céline S. Goçkalves^{1,2}*

¹Life and Health Sciences Research Institute (ICVS), School of Medicine, University of Minho, Braga, Portugal

²ICVS/3B's—PT Government Associate Laboratory, Braga/Guimarães, Portugal

³Medical Devices, International Iberian Nanotechnology Laboratory – INL, Braga, Portugal

⁴Department of Neurosurgery, Hospital Braga, Braga, Portugal

⁵Department of Oncology, Hospital Braga, Braga, Portugal

*Presenting author

Introduction: Glioblastoma (GBM) is the most aggressive primary brain tumor, characterized by poor prognosis and limited treatment options. Our team showed that WNT6, a Wnt signaling ligand, is overexpressed in GBM and associated with tumor aggressiveness. Notably, WNT6 was highly enriched in extracellular vesicles (EVs) derived from GBM primary cells. Given the role of EVs in intercellular communication and their potential as minimally invasive liquid biopsy tools, this work explores the biological and clinical relevance of WNT6-driven EV signaling in GBM.

Methods: GBM cell lines with genetically altered WNT6 expression were used. Additionally, plasma from GBM (n=20) and lower grade glioma (n=14) patients were collected from Hospital de Braga. EVs were isolated from conditioned medium (size exclusion chromatography) and plasma (precipitation) and characterized by nanoparticle tracking analysis and western blot. Transcriptomic (RNA-seq) and proteomic (MS/MS) profiles of cells and EVs were assessed by differential expression and pathway analyses, including GO enrichment and GSEA. WNT6 levels in EVs were quantified by western blot, ddPCR and ELISA.

Results: RNA-seq and MS/MS analyses of WNT6-high cells revealed enriched EV and exosome-related signatures. Consistently, WNT6-high cells released more EVs per cell compared to WNT6-low. This effect was reflected in patient samples, where plasma EV concentration was increased in WNT6-high tumors, highlighting a link between WNT6 and EV biogenesis. Furthermore, WNT6 reshaped EV transcriptomic and proteomics cargo, with enrichment in oncogenic pathways. Interestingly, WNT6 mRNA and protein were detected in EVs from GBM cells and patient plasma, closely mirroring its cellular and tumoral expression levels, respectively.

Conclusion: WNT6 modulates EV release and shapes oncogenic EV cargo in GBM. Additionally, its detectability in circulating EVs, reflecting tumor expression levels, supports its potential as a liquid biopsy biomarker in glioma.



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